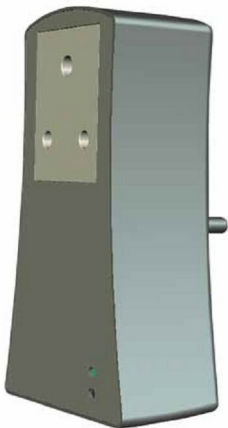


A 'SMART' PLUG THAT MONITORS POWER CONSUMPTION AND GENERATES ALERTS, WIRELESSLY!

Smart Plugg is a 'smart' energy meter, developed to wirelessly monitor power consumption and notify users when it observes drastic changes in the energy consumption pattern



Smart Plugg

■ JALAJA RAMANUNNI

Often we do not realise that homes and industries consume much more energy than required due to inefficient electronics, faulty devices and our own negligence. This is especially true in sectors such as hospitality, healthcare and networking

devices at homes. Studies prove that we can save at least five to fifteen per cent electricity at homes and in industries by using smart meters to manage electricity demands. Imagine the difference smart meters can make in the long run.

To curb this problem, a plug-and-play energy meter named Smart Plugg has been developed by Suvana Solutions, Secunderabad. The device is a combination of a smart meter and data logger. It has a built-in wireless communication system developed for electricity-usage control. It not only records the consumption of electric energy in real time, but also communicates this information immediately in the form of an alert to the user when a device starts consuming more energy than usual.

Need for neat connections

Suvana Solutions started out with energy meters which displayed water, gas and electricity consumed at homes. Out of these, electricity was the most expensive and people felt a need to monitor its usage. Though there are many smart metering devices available in the market, Suvana Solutions felt the need for a wireless metering device.

"Normal energy meters can read the amount of energy consumed, but are not capable of notifying owners immediately about a drastic difference in power consumption. This is the sole reason that made us feel there was need for wireless energy meters. Moreover, everyone is shifting to neat

connections with the help of wireless technology as it is difficult to lay out cables in already existing homes and organisations," affirms Phani Varanasi, founder and director, Suvana Solutions. Varanasi claims this is the first energy meter in India which offers wireless communication technology.

Smart Plugg also allows home and building owners to monitor the amount of electricity being used by individual appliances and set thresholds on energy consumption. It sends alerts to users if the energy consumption exceeds normal limits to cut-off power remotely from the appliance.

How it works?

You can easily plug-in the Smart Plugg into a normal power socket, and run your appliance through it. Any electrical appliance or device that uses a standard Indian 16A plug can be monitored with the Smart Plugg.

The device constitutes two sets of chipsets. The metering is based on an Analog Devices' system-on-chip single-phase metering IC, and the wireless connectivity is established using Freescale chipset-based ZigBee communication. It measures how much electricity your appliance consumes the first time and wirelessly communicates this information to a PC, mobile phone or building management system anywhere in the world through an Internet-linked Ethernet gateway, where this information gets stored.